



Semester: August 2022-December 2022		
Maximum Marks: 30	Examination: In-Semester Examination	Duration : 1Hr
Programme code: 01	Class: SY	Semester: III
Programme: B. Tech Computer Engineering		(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer	
Course Code: 116U01C301	Name of the Course: Integral transform and Vector calculus	

Question No.		Max. Marks
Q.1	Attempt any THREE of the following	
(a)	Find the Laplace transforms of $\int_0^t \frac{1-e^{au}}{u} du$	05
(b)	Evaluate $\int_0^\infty e^{-3t} t \cos t \cdot dt$ using Laplace transforms.	05
(c)	Find the inverse Laplace transforms of $\left(\frac{1}{s^2-s-6}\right)$ by using convolution theorem.	05
(d)	Find the inverse Laplace transforms of $\log\left(1 + \frac{1}{s^2}\right)$	05
(e)	Find the Laplace transforms of $(1 + 2t - t^2 + t^3)H(t - 1)$	05
Q.2 (a)	Obtain Fourier series for $f(x) = x \cos x$ in $(-\pi, \pi)$ OR Obtain complex form of Fourier series for $f(x) = e^x$ in $(-1, 1)$	05
(b)	Obtain the Fourier expansion of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in $0 \leq x \leq 2\pi$ and $f(x + 2\pi) = f(x)$. Also deduce that (i) $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ (ii) $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ (iii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$ (iv) $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$	10



Semester: August 2022-December 2022

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Examination: In-Semester Examination

Duration :1 hr:15 min

Programme code: 116U01C305

Programme: B Tech Computer Engineering

Class: SY

Semester: III
(SVU 2020)

Name of the Constituent College:

K. J. Somaiya College of Engineering

Name of the department:
COMP

Course Code: 116U01C303

Name of the Course: Discrete Mathematics

Question No.		Max. Marks
Q 1	1) Determine the number of integers between 1 and 250 that are divisible by any of the integers 2, 3, 5 and 7. OR In a survey, it is reported that of 1000 programmers, 650 habitually flowchart their programs, 788 are skilled Python programmers, 675 are men, 278 of the women are skilled Python programmers, 440 programmers habitually flowchart and are skilled Python programmers, 210 habitually flowchart and 166 women are both skilled in Python and habitually flowchart. Would you accept these data as being accurately reported? Justify your answer.	10M (6M+4M)
Q 2	Find whether the following statement is tautology? $(p \Leftrightarrow q) \Rightarrow (\neg p \wedge r)$	
Q2	Solve any two. (I) Show that $1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$ (II) Find out CNF of the following statement. (Using laws of logic or using truth table). $(p \Leftrightarrow q) \Rightarrow (\neg p \wedge r)$ (III) Find whether the relation is equivalence relation? $A = \{1, 2, 3, 4, 5\} = B$ aRb iff a is multiple of b. Write the relation.	5 M 5 M
Q3	Consider the relation with $A = \{1, 2, 4, 6, 8, 10\} = B$ such that, aRb iff a divides b. Calculate R R^2 R^3 R^∞ Represent R, R^2, R^3 relations with digraph.	10M



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Programme code:

Programme: Computer Engineering

Class:

FY/SY/TY/LY/MTECH

Semester:

**I/H/III/IV/V/VI/VII/VIII
(SVU 2020)**

Name of the Constituent College:

K. J. Somaiya College of Engineering

Name of the department:

COMP/ETRX/EXTC/IT/MECH

Course Code:116U01C304

Name of the Course: Object Oriented Programming Methodology.

Question No.		Max. Marks
Q.1	Explain what is Auto-boxing and Unboxing with examples.(05M) Write the differences between Structural and Object Oriented Programming approaches.(05M)	10 M
Q.2	Write a program which accepts information about n no of students from user (10 M) Create an array of objects to store stud_id , name, avg_marks. Your program should provide following functionalities 1. To add students according to the descending order of their average marks. 2. To delete any student data based on the student id. 3. To display student details according to the descending order of their Average marks.	10 M
Q.3 (A)	What is the significance of constructor? Explain types of Constructor with examples in detail. (05M) OR Explain Method overloading and Method Overriding with examples. (05M)	05 M
Q.3 (B)	During shopping, when you supply your credit card number to an online vendor, data input errors are checked before your credit card is validated. For example, credit cards issued by Visa all have numbers beginning with the digit 4, and those issued by American Express begin with 34 or 37. Another approach of validation is the Luhn algorithm which detects some, but not all, invalid numbers. This algorithm can alert a vendor to some bad numbers. The method works as follows: Beginning with the second-rightmost digit and moving right to left, double every other digit. If the doubling process produces a value greater than 9, subtract 9 from that value. For example, to check the validity of credit card number	05 M

5113 4765 1234 8002 proceed as follows:

Double alternate digits. Subtract 9 from products exceeding 9.

	5	1	1	3	4	7	6	5	1	2	3	4	8	0	0	2
	↓		↓		↓		↓		↓		↓		↓		↓	
Doubled digits →	10	-9=1	2		8	12-9=3	2		6	16-9=7	0		0			

Write a program that determines whether a credit card number with 16 (or fewer) digits passes the Luhn test. (05M)

OR

Given a string consisting of lower case English alphabets, Write a program to find the number of distinct subsequences of the string. (05M)

Example :

Input: $s = \text{"gfg"}$

Output: 7

Explanation: The seven distinct

subsequences are "", "g", "f", "gf", "fg",

"gg" and "gfg"



SOMAIYA
VIDYAVIHAR UNIVERSITY

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Programme: B Tech Computer Engineering		(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP	
Course Code: 116U01C303	Name of the Course: Computer Organization and Architecture	

Question No.		Max. Marks
Q1	Draw the basic and detail structure of Von Neumann model and state each component. OR Write a short note on SCSI bus. Support your answer with a neat diagram. Also comment on various SCSI bus versions.	10M
Q2	Is Booth's recording better than Booth's algorithm? Justify your answer. Hence solve using Booths recoding clearly stating each step M= 9 Q= -15	10M
Q3	Identify and justify addressing modes in the following examples 1. ADD [1234h],AX 2. MOV AX,[BX+05h] 3. ADD AX, #30h 4. ADD [BX],AX 5. MOV AX,CX	10M



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Class: SY

**Semester: III (SVU
2020)**

Programme: B. Tech

Name of the Constituent College:

K. J. Somaiya College of Engineering

Name of the department:

COMP/IT

Course Code: 116U01C302

Name of the Course: Data Structures

Question No.		Max. Marks
Q1	Solve Any TWO a. Differentiate between linear data structure and Non linear Data structure. b. Comment on significance of ADT. c. Discuss ADT as a concept with an example.	10
Q2	Write a Pseudocode/algorithm for implementation of the following operations on singly linked list. (Consider all possible cases) i. Insertion in Between ii. Searching a Data item in a Linked List OR <u>Suggest and justify</u> a suitable data structure for the following problem definitions. . Illustrate with a suitable example. i. Consider a music application which keeps track of songs played maximum number of times. Based on this frequency and recency, the application creates a suitable playlist for the user. The data structure should maintain the suggested playlist. ii. Consider a multiplayer game: "passing the pillow". When the music stops, the participant with the pillow has to perform an activity, gets eliminated and the game continues with the rest of them. The data structure should maintain the list of the participants, keep track of who's eliminated, remaining list and can announce the final winner.	10
Q3	Convert the given infix expression into postfix using stack. Show the contents of stack and output string with every input element. $a+b-c/d^f*m*n+k+p$	10